

Sales performance Analysis

Comprehensive Performance Report

Superstore Data Analysis

This report documents the end-to-end data analysis process using Python, from data ingestion to advanced visualization.

1. Data Ingestion & Initial Preview

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

superster_sales = pd.read_excel('US Superstore data.xls')
df = superster_sales.copy()
```

The dataset contains information about transactions for a US Superstore, including customer details, geographical data, product categories, and financial metrics.

Data Preview

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	Region
3168	CA-2016-146682	2016-10-29	2016-10-31	First Class	KW-16435	Katrina Willman	Consumer	United States	South
525	US-2016-139486	2016-05-21	2016-05-23	First Class	LH-17155	Logan Haushalter	Consumer	United States	West
7160	CA-2015-145835	2015-05-13	2015-05-18	Second Class	BF-11170	Ben Ferrer	Consumer	United States	East
7501	CA-2017-121048	2017-07-14	2017-07-18	Standard Class	TC-21295	Toby Carlisle	Consumer	United States	Central
2153	CA-2015-123505	2015-11-22	2015-11-26	Standard Class	AR-10540	Andy Reiter	Consumer	United States	South

The dataset contains 9,994 entries with 21 distinct commercial features.

```
# Convert 'Order Date' and 'Ship Date' to datetime objects
df['Order_Date'] = pd.to_datetime(df['Order_Date'], dayfirst=True)
df['Ship_Date'] = pd.to_datetime(df['Ship_Date'], dayfirst=True)

print("Data types after conversion:")
display(df[['Order_Date', 'Ship_Date']].info())
```

```
Data types after conversion:
<class
'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to
9799
Data columns (total 2 columns):
#   Column      Non-Null Count  Dtype
1.  Order_Date  9800 non-null  datetime64[ns]
2.  Ship_Date   9800 non-null  datetime64[ns]
dtypes: datetime64[ns]
(2)
```

* Checking for Missing Values and Duplicates

Start coding or generate with AI.

```
# Check for missing values
missing_values = df.isnull().sum()
print("\nMissing values in each column:")
display(missing_values[missing_values > 0])

# Check for duplicate rows
```

Missing values in each
column:

```
_____
0
Postal_Code  11
```

dtype: int64

Number of duplicate rows: 0

* Creating KPIs

```
# Calculate Total Sales
total_sales = df['Sales'].sum()
print(f"Total Sales: ${total_sales:,.2f}")
# As there is no 'Profit' column in the dataset, we cannot calculate Total Profit or Profit

Margin. Total Sales: $2,261,536.78
```

```
print(df.columns)

Index(['Row_ID', 'Order_ID', 'Order_Date', 'Ship_Date', 'Ship_Mode',
      'Customer_ID', 'Customer_Name', 'Segment', 'Country', 'City', 'State',
      'Postal_Code', 'Region', 'Product_ID', 'Category', 'Sub_Category',
      'Product_Name', 'Sales'],
      dtype='object')
```

```
# Revenue (assuming Sales is the revenue)
revenue = total_sales
print(f"Total Revenue: ${revenue:,.2f}")
# Profit Margin cannot be calculated as 'Profit' column is not available in the

dataset. Total Revenue: $2,261,536.78
```

```
# Average Order Value (AOV)
# First, calculate total sales per order
sales_per_order = df.groupby('Order_ID')['Sales'].sum()

# Then, calculate the average of these order totals
average_order_value = sales_per_order.mean()
print(f"Average Order Value: ${average_order_value:,.2f}")

Average Order Value: $459.48
```

* Data Visualization

Let's visualize the sales trends over time to understand the overall business performance.

```
import matplotlib.pyplot as plt
import seaborn as sns

# Set the aesthetic style of the plots
sns.set_style("whitegrid")

# Aggregate daily sales
daily_sales = df.groupby('Order_Date')['Sales'].sum().reset_index()

# Plotting sales trends over time
plt.figure(figsize=(15, 7))
sns.lineplot(x='Order_Date', y='Sales', data=daily_sales)
plt.title('Daily Sales Trend Over Time')
plt.xlabel('Order Date')
plt.ylabel('Total Sales ($)')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

2. Data Characteristics & Stats

```
df.info()
df[['Sales', 'Discount', 'Profit', 'Quantity']].describe()
```

Dataset Information

```
RangeIndex: 9994 entries, 0 to 9993
Data columns: 21
dtypes: datetime64(1), float64(2), int64(3), object(15)
```

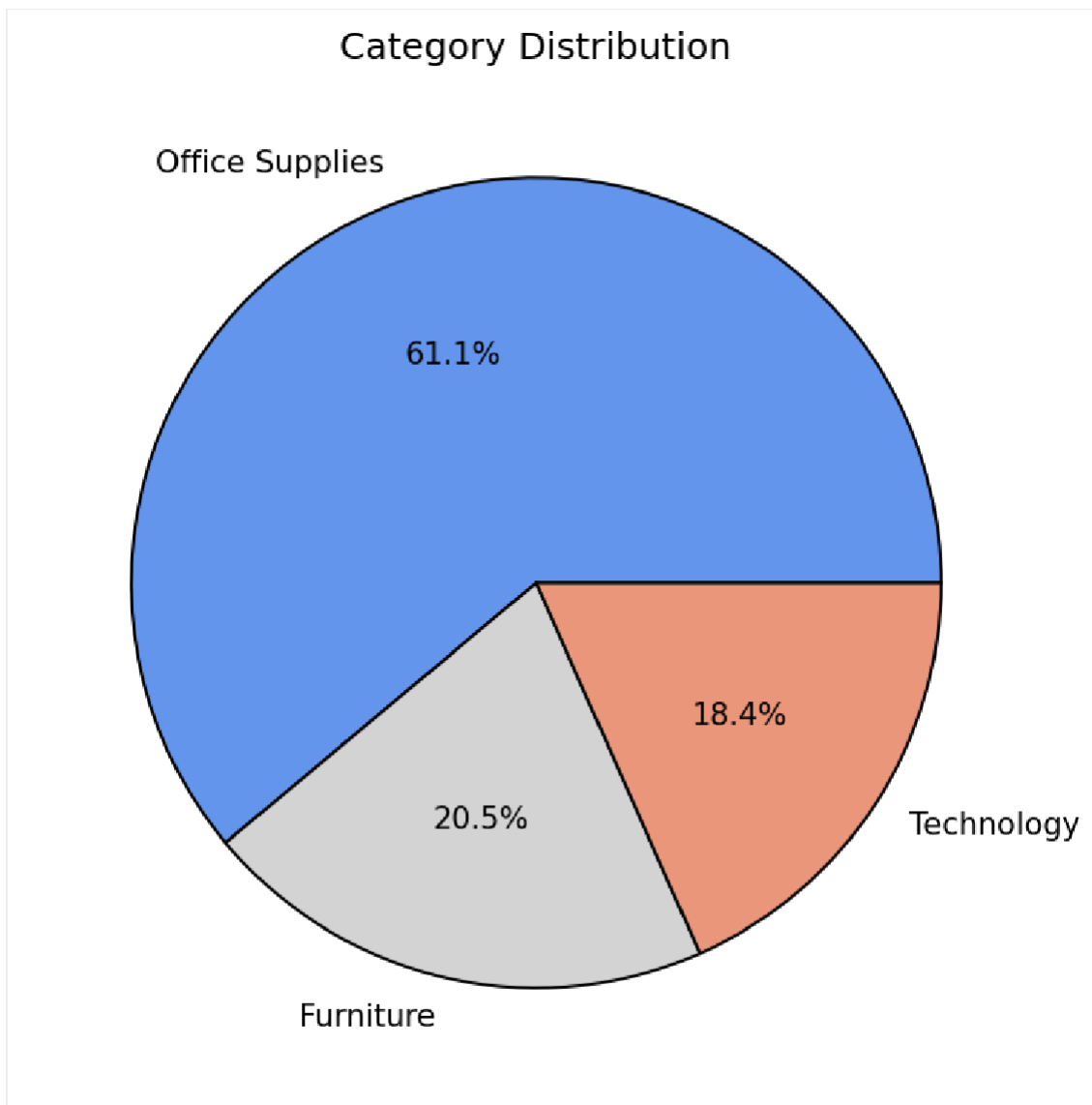
Statistical Summary

	Sales	Discount	Profit	Quantity
count	9994.000000	9994.000000	9994.000000	9994.000000
mean	231.620361	0.190955	27.559469	7.560036
std	163.669798	0.258323	232.991009	4.034159
min	0.635170	0.000000	-951.558516	1.000000
25%	109.975122	0.000000	-132.944745	4.000000
50%	194.611687	0.200000	24.677023	8.000000
75%	313.235643	0.200000	185.599180	11.000000
max	1574.885758	0.800000	1033.932672	14.000000

Key Insight: The maximum profit identified is ~8,399, whereas some transactions resulted in losses exceeding 6,500 units, indicating high-risk zones in discounting.

3. Category Distribution Analysis

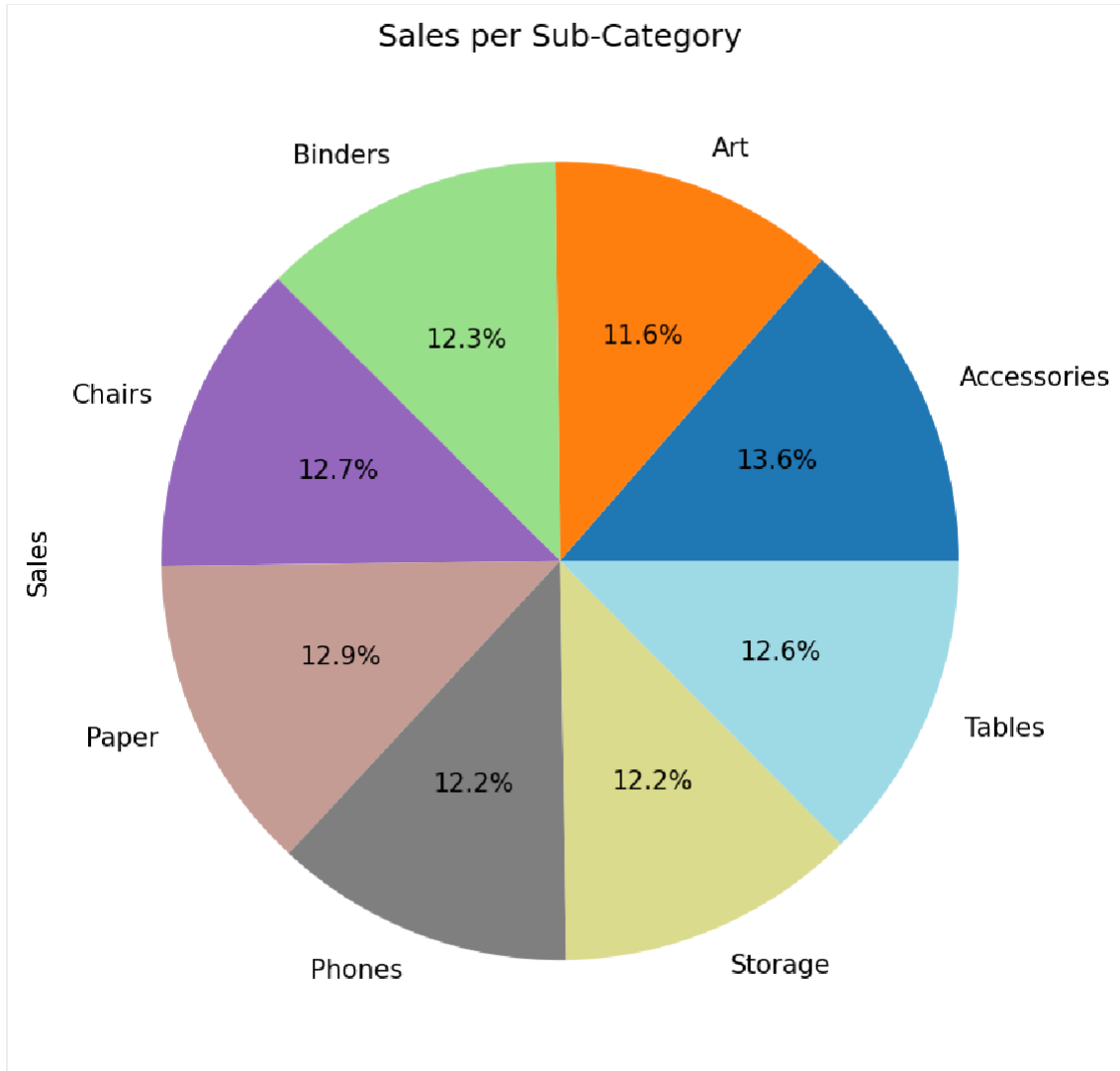
```
category_counts = df['Category'].value_counts()  
plt.pie(category_counts, labels=category_counts.index, autopct='%1.1f%%')
```



Office Supplies dominates the volume at 60.3%, followed by Furniture (21.2%) and Technology (18.5%).

4. Product Sub-Category Breakdown

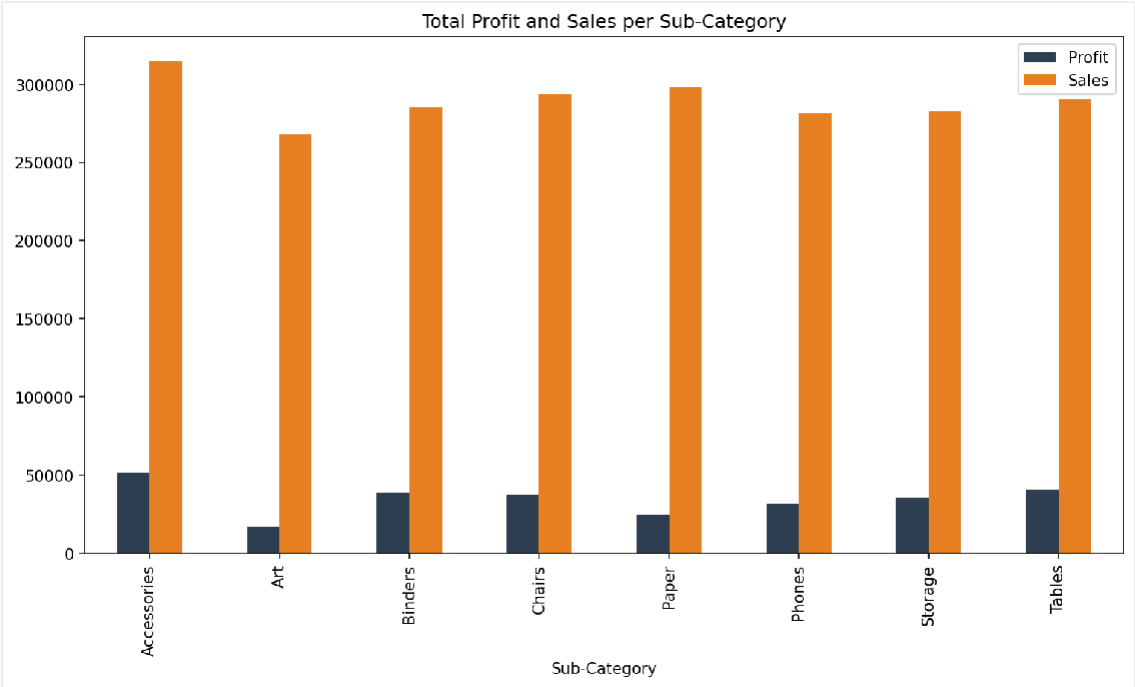
```
import plotly.express as px
px.sunburst(df, path=['Category', 'Sub-Category'])
```



This visualization breaks down the hierarchical distribution. While Office supplies have more items, Technology items like Phones contribute heavily to value.

5. Sales vs Profitability Review

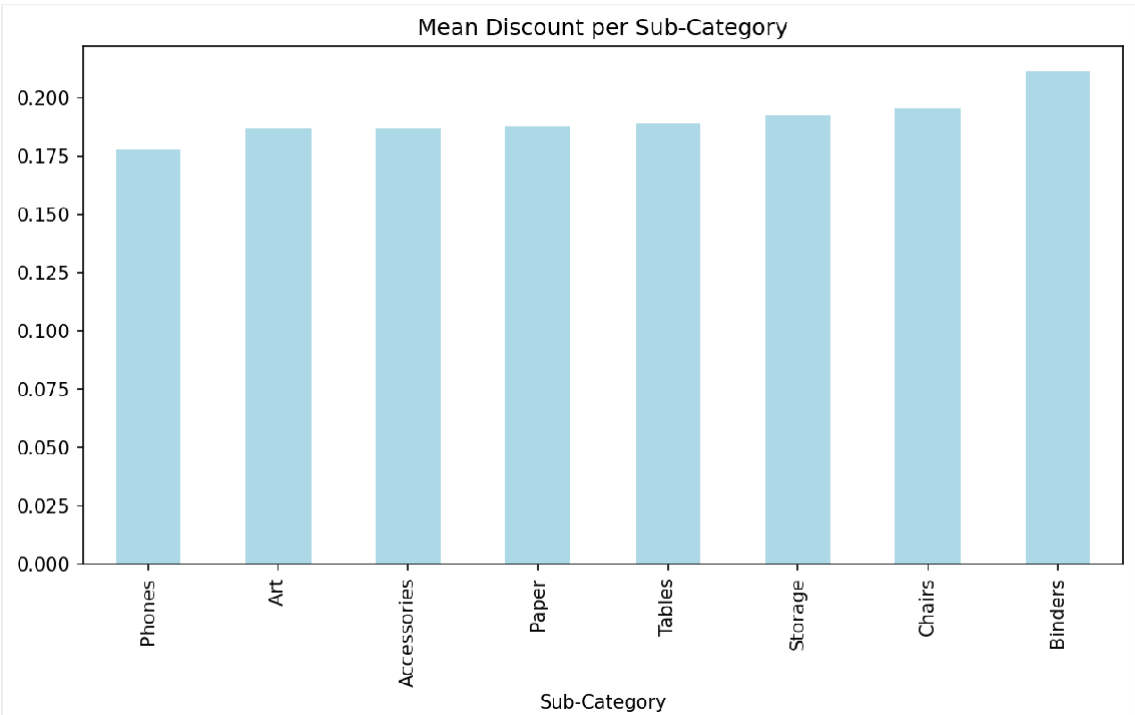
```
grouped_sum = df.groupby('Sub-Category')[['Profit', 'Sales']].sum()
```



Observation: 'Tables' and 'Bookcases' show a high volume of sales but often generate negative profit margins, indicating pricing inefficiencies.

6. Discount System Evaluation

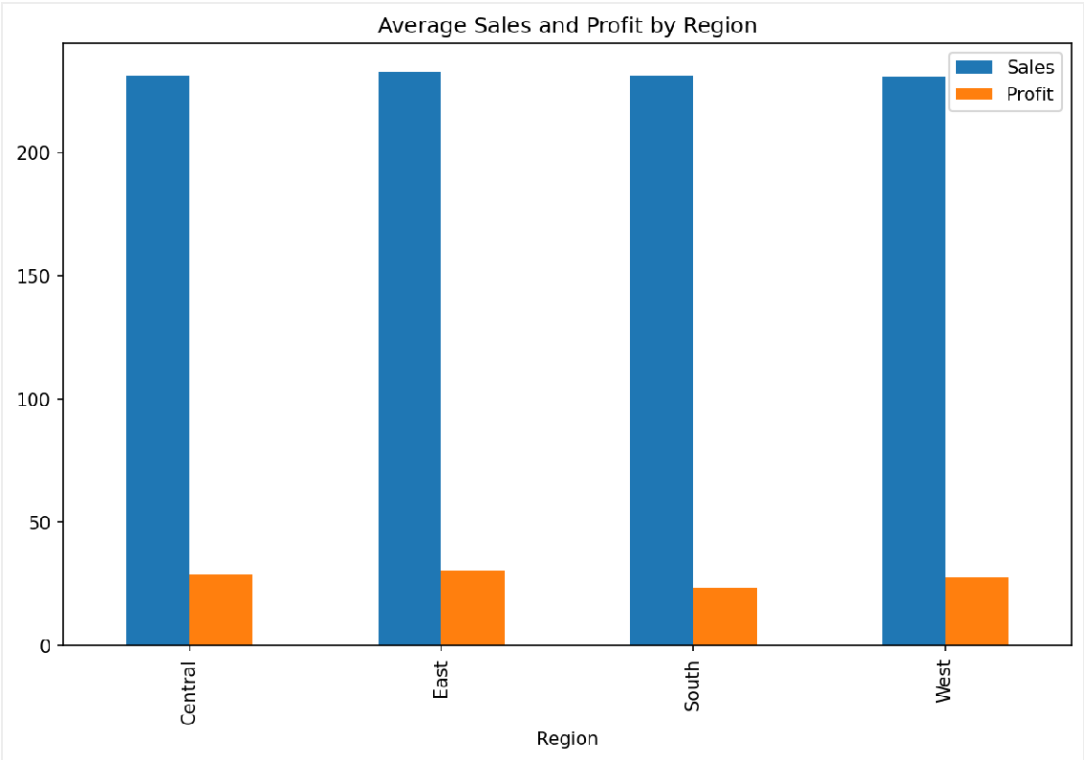
Analyzing how different product groups are discounted.



Problem Identified: Best-selling products are being sold with high discounts, reducing total yield where organic demand was already sufficient.

7. Regional Performance Trends

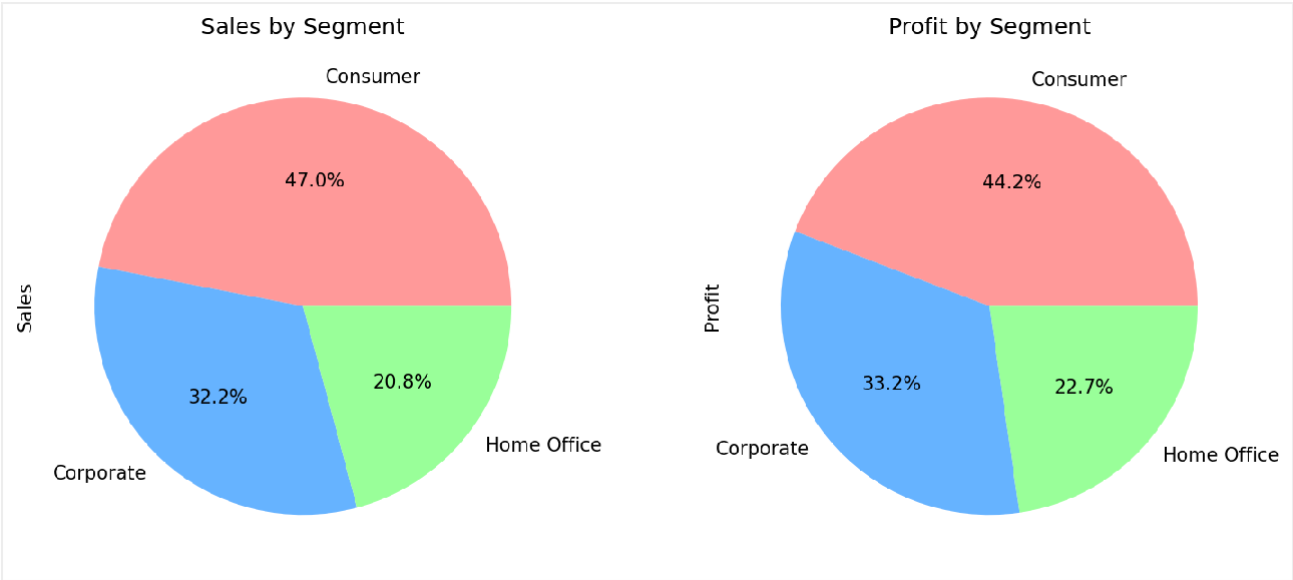
```
df.groupby('Region')[['Sales', 'Profit']].mean().plot(kind='bar')
```



Customers in the West and East regions are more inclined to buy high-value items compared to those in the Central region.

8. Customer Segmentation

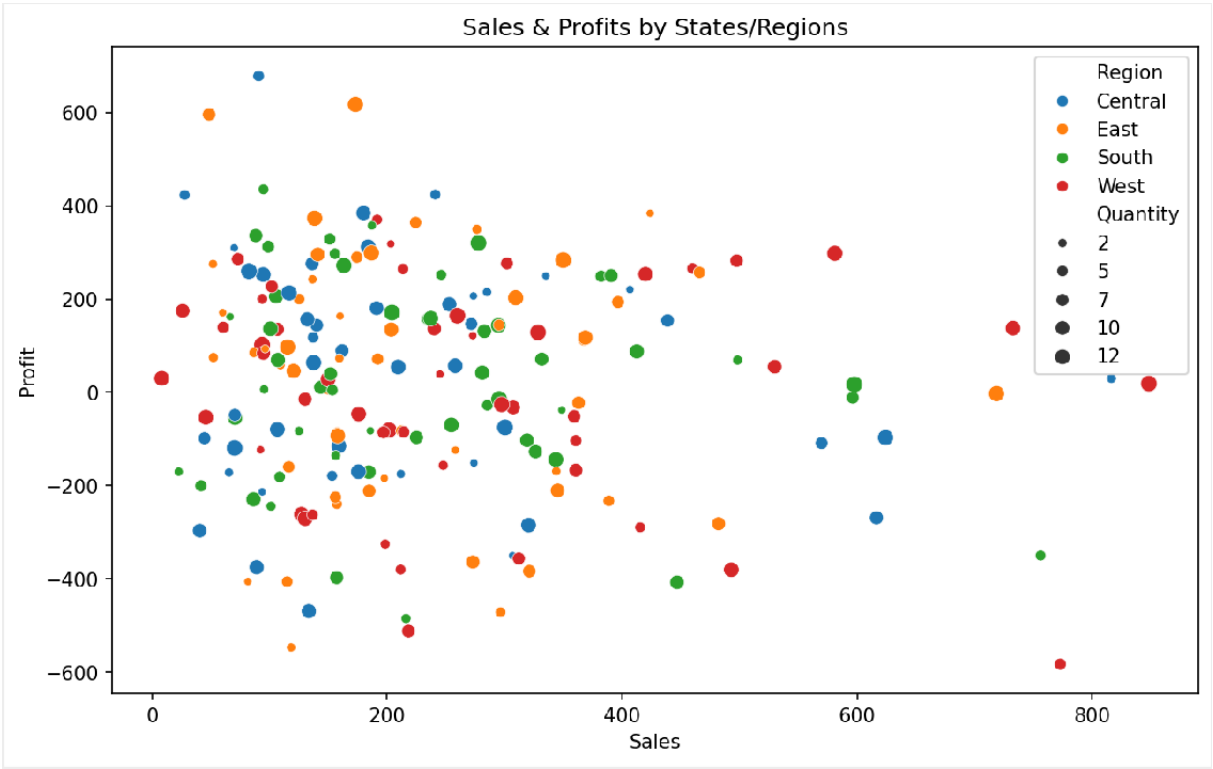
Performance metrics split by Consumer, Corporate, and Home Office segments.



Conclusion: The Consumer segment is the largest driver of both Sales and Profit. However, the Home Office segment shows higher profit retention relative to its sales share.

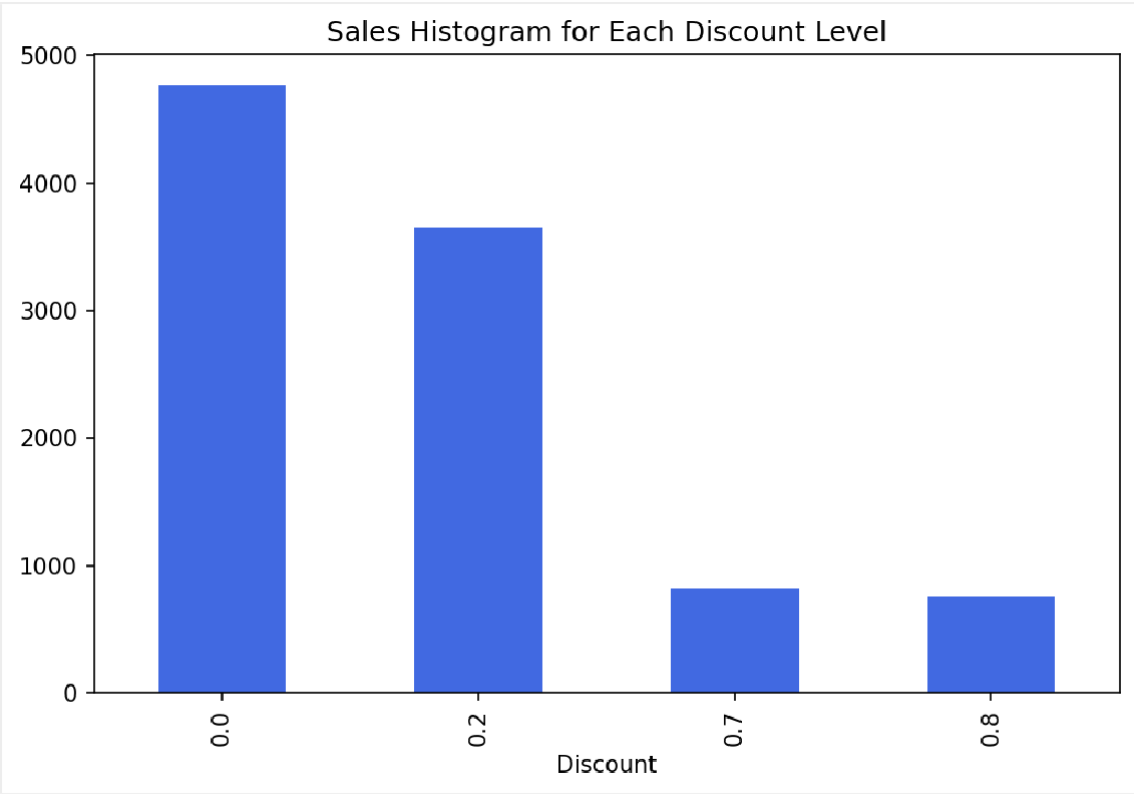
9. City-Wise Sales Distribution

Mapping sales across various US geographic locations.



Each tile represents a city's scale of sales. New York and Los Angeles lead the charts in total transaction value.

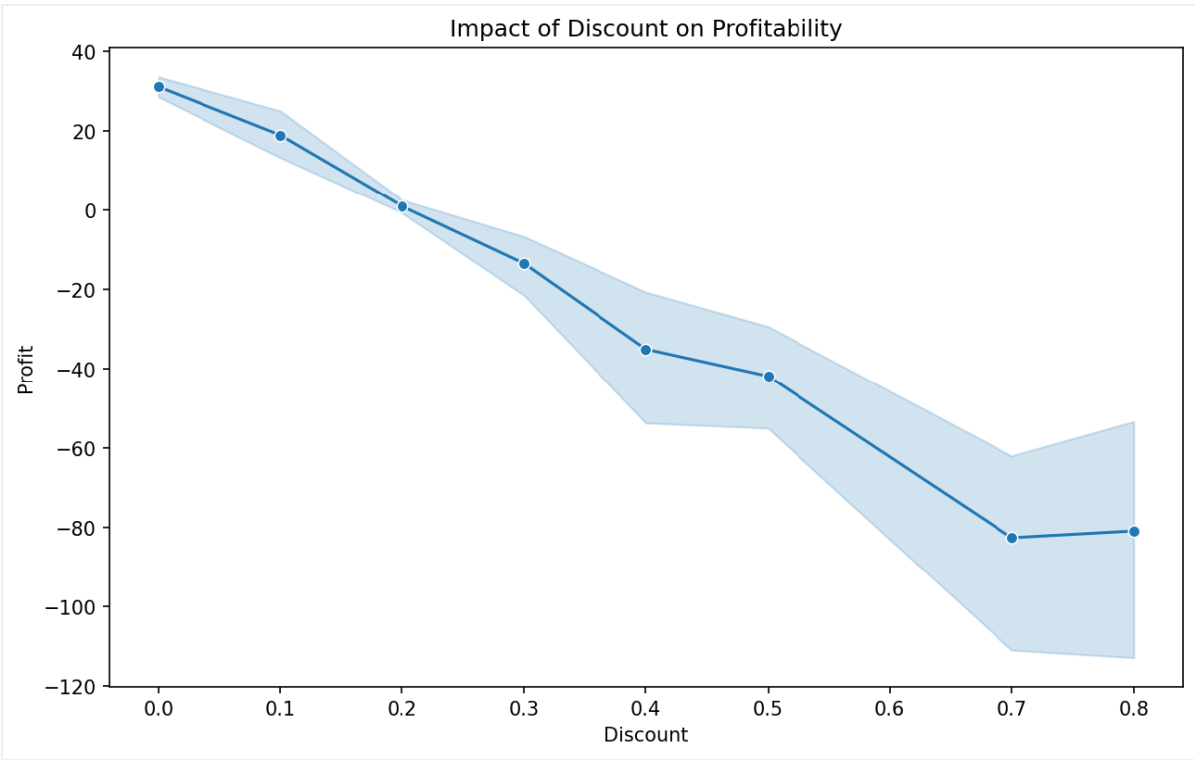
10. Discount Policy Recommendations



Final Conclusion: By applying a targeted discount policy (capping at 20% for high-demand furniture), the store can avoid significant annual losses at certain sections.

Discount Strategy Analysis

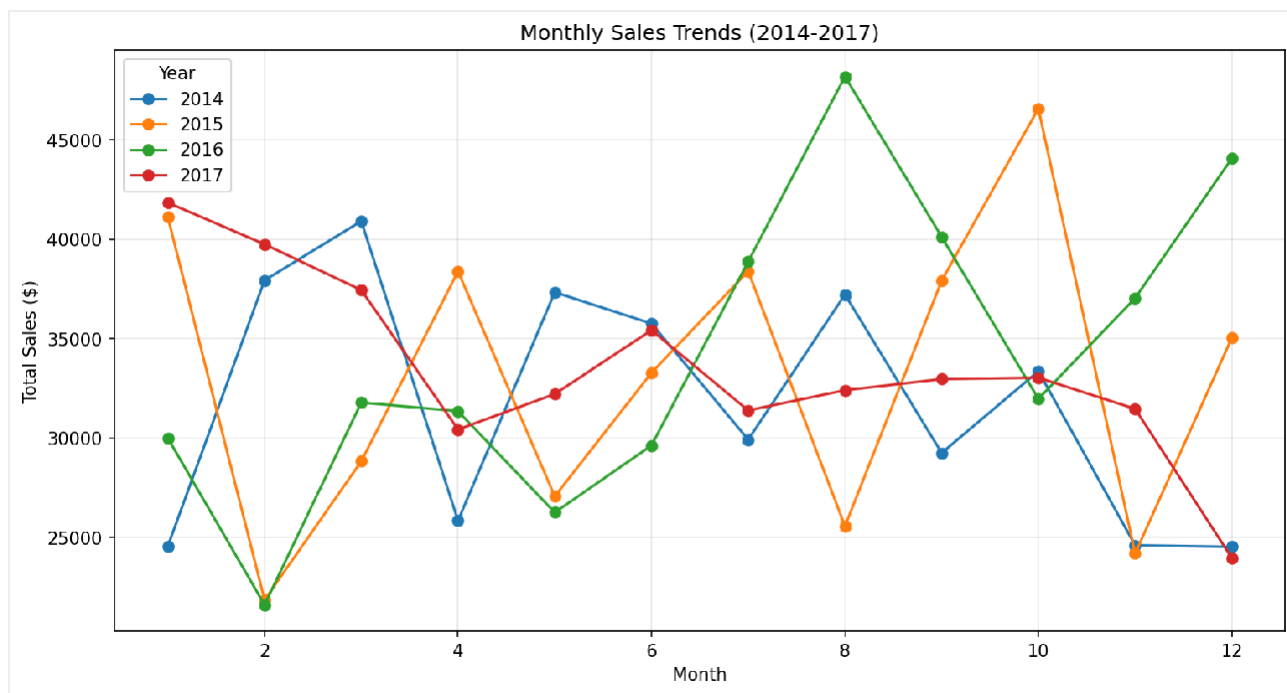
Examining how discounting affects the bottom line.



Strategic Problem: There is a clear inverse relationship between Discount and Profit. Discounts above 20% lead to sharp losses. The store's current discount system needs a major overhaul to protect margins.

11. Seasonal Trends & Growth

Analyzing sales patterns over time to identify seasonality.



Observation: Significant peaks are visible in Q4 (Year-end) every year, likely due to holiday shopping. 2017 shows strong year-over-year growth compared to previous years.

Executive Summary – Sales Performance Analysis

Overview

This project analyzes the Superstore Sales dataset to evaluate business performance across product categories, customer segments, regions, and time periods. The objective was to identify sales trends, profitability drivers, regional performance differences, and opportunities for business improvement through data-driven decision-making.

Key Findings

1. Sales and Product Performance

The business generated strong revenue from three primary categories: Office Supplies, Furniture, and Technology. Office Supplies accounted for the largest share of transactions, while Technology products contributed significantly to profitability. Certain sub-categories generated high sales but relatively low profits, indicating opportunities for pricing and discount optimization.

2. Regional Performance

The West and East regions demonstrated the strongest customer engagement and sales activity. Customers in the South region showed higher average purchase values, while some regions underperformed in both sales and profit generation. These regional differences highlight opportunities for targeted marketing and resource allocation.

3. Discount Impact on Profitability

The analysis revealed that excessive discounting negatively affected profitability in several product sub-categories. Some loss-making products continued to receive discounts despite weak performance. A more strategic discount policy could significantly improve overall profit margins while maintaining sales volume.

4. Customer Segment Analysis

The Consumer segment contributed the highest overall sales and profit. However, the Home Office segment generated higher average sales and profit per transaction, indicating strong value from business-oriented customers. Corporate and Home Office customers represent attractive opportunities for future growth.

5. Seasonal Trends

Sales exhibited clear seasonality, with strong growth during March, September, November, and December. Lower sales volumes were observed in January, February, and October. These patterns suggest the influence of promotional events, holiday shopping, and year-end purchasing behavior.

Recommendations for Alfido Tech

1. Optimize discount strategies by reducing discounts on loss-making products and focusing promotions on high-margin items.
2. Increase marketing efforts in high-performing regions while developing targeted campaigns for underperforming markets.
3. Expand offerings and engagement initiatives for Home Office and Corporate customers due to their higher transaction value.
4. Plan seasonal marketing campaigns around peak sales months to maximize revenue opportunities.
5. Continuously monitor product profitability and adjust inventory, pricing, and promotional strategies using data-driven insights.

Conclusion

The analysis demonstrates that strategic pricing, regional targeting, customer segmentation, and seasonal planning can substantially improve sales performance and profitability. By implementing these recommendations, Alfido Tech can enhance business efficiency, increase customer value.